Tetris 2.0: Test Plan

The four required test types, with the plan and files for each. All four are in place.

Run everything from the game folder:

```
pip install pygame
python3 -m unittest test_tetris_render test_tetris_auth test_tetris_features test_tetris_cloud test_tetris_extras test_tetris_regression test_tetris_acceptance
```

Unit tests

Plan/files: test each rule and helper on its own, with no window and no network, so a failure points to one place.

File	Tests	Covers
`test_tetris_auth.py`	56	Username and password rules, hashing, registering, signing in, file loading, score paths
`test_tetris_features.py`	43	Pause, stats, combos, back-to-back, level up, T-spins, lock delay, themes, settings, sound generation
`test_tetris_render.py`	10	Score file load and save, drawing helpers, ghost piece

Integration tests

Plan/files: test that the parts work together: the real score server with the real client over HTTP, the game with the score files, and the play loop with fake key events.

File	Covers
`test_tetris_cloud.py`	Server on a free port with the real client: keys, bad input, saving across a restart, merging local and cloud scores, game over sending the score to the server
`test_tetris_features.py` (`PlayLoopSmokeTest`)	Pause, theme, mute, game over, and restart through the real play loop
`test_tetris_extras.py`	B key reaching the bomb, music following game state

Regression tests

Plan/files: make sure changes don't quietly break what already works. `test_tetris_regression.py` has three parts.

Part	What it does
Golden values	Locks the scoring tables, gravity curve, lock delay, bomb limits, account limits, and music tempo. Changing one on purpose means updating the test
Past trouble spots	Damaged files, usernames with path tricks, zero scores, double counting in sync, pause freezing everything, bombs on an empty floor, hold rules, unknown sounds, passwords never stored
Random play	25 seeded random games of 1,500 moves each. After every move: the piece is in a valid spot, the score never drops, the queue has 3 pieces, bombs stay between 0 and 3, the level matches the lines cleared

Acceptance tests

Plan/files: check that the game does what the requirements say, written as Given / When / Then stories from the player's point of view. `test_tetris_acceptance.py` has 20, named after requirements R1 to R10.

Group	Stories
Accounts (R1)	New player signs up; returning player signs in; wrong password and unknown name give the same message; sign up explains what to fix; passwords are never written down
Gameplay (R2 to R6, R8)	Piece falls and can be steered; hard drop lands and the next piece appears; 3-piece preview and one hold per piece; a Tetris scores 800; 10 lines raise the level; a Tetris earns a bomb and the bomb clears rows; pause and resume; game over with stats
Saved data	Each player keeps their own top 5; theme and sound are remembered

Results Log

Date	Tester	Unit	Integration	Regression	Acceptance	Notes